

Plant Defense Against Herbivores: Chemical Aspects

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Abstract

Plants have evolved a plethora of different chemical defenses covering nearly all classes of (secondary) metabolites that represent a major barrier to herbivory: Some are constitutive; others are induced after attack. Many compounds act directly on the herbivore, whereas others act indirectly via the attraction of organisms from other trophic levels that, in turn, protect the plant. An enormous diversity of plant (bio)chemicals are toxic, repellent, or antinutritive for herbivores of all types. Examples include cyanogenic glycosides, glucosinolates, alkaloids, and terpenoids; others are macromolecules and comprise latex or proteinase inhibitors. Their modes of action include membrane disruption, inhibition of nutrient and ion transport, inhibition of signal transduction processes, inhibition of metabolism, or disruption of the hormonal control of physiological processes. Recognizing the herbivore challenge and precise timing of plant activities as well as the adaptive modulation of the plants' metabolism is important so that metabolites and energy may be efficiently allocated to defensive activities.

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INTRODUCTION: PRINCIPLES OF PLANT DEFENSE

Throughout their entire life cycle, higher plants are challenged by many different abiotic and biotic stresses. Biotic stress is represented, in particular, by heterotrophic organisms, which all depend on the energy fixed by autotrophic plants. Hence, heterotrophic organisms try everything to use plants as a food source. As sessile organisms, plants have no chance of escaping attacks from organisms, so they must employ other strategies to defend themselves. Numerous strategies are based on the tremendous diversity within plant chemistry, e.g., the ability to synthesize more than 200,000 estimated compounds, referred to as specialized metabolites, that evolved in response to particular ecological challenges (96). Besides phytopathogenic microorganisms, herbivorous insects and other arthropods must also be defended against. Among these are specialists that feed on only a limited number of plant species, or even one single host, and generalists that can feed on numerous species. Because plants and insects have coexisted for at least 350 million years, plants have developed

successful defensive traits (47), many of which may have been involved in plant-microbe interactions millions of years before. In principle, two broad categories of plant defenses can be distinguished: (*a*) always present and (*b*) inducible, which may be specifically elicited by certain aggressors. For instance, a chewing caterpillar (**Figure 1**) can cause different defense reactions than can a cell-sucking spider mite (74). For such efficient discrimination, plants must be able to recognize herbivores with a high degree of sophistication in combination with intracellular signaling and conversion of those signals into appropriate biochemical, physiological, and cellular responses (79, 80). In almost all cases, upon herbivore attack, an inducible defense is established locally on the site of infestation as well as systemically throughout the whole plant, albeit in some cases with lower intensities (85).

A further distinction can be made between both constitutive and inducible defenses: Each can be either direct or indirect (**Figure 2**). Direct defenses act by themselves against the aggressor. Typical examples are morphological features such as thorns, prickles, or high levels of lignification. Trichomes may fulfill both features: They are a mechanical barrier, but glandular trichomes may harbor secretory structures that contain feeding or egg-deposition deterrents as well as toxins (41); however, probably more important are the specialized metabolites of various tissues, which can be toxic, antidigestive, or, at least, unpalatable. Indirect defenses act via the attraction of organisms from an additional trophic level, e.g., of enemies of the attacking herbivores (53). The release of certain volatile organic compounds (VOCs), consisting mainly of terpenoids, fatty acid derivatives, and a few aromatic compounds, by herbivore-infested plants, for example, can attract parasitoids, in particular, or predators of the feeding insect (33, 38, 39, 66). Many VOC blends are produced “on demand” after mechanical or biological challenge, and their composition depends on the mode of damage, such as wounding (86), egg deposition (56), and herbivore feeding. The insect feeding-induced

Toxicity: general term for the property of substances indicating the degree to which a compound can damage a cell or organism; it is dose dependent and measured by the effect on the particular target

Trophic level: the position occupied by an organism in a food chain

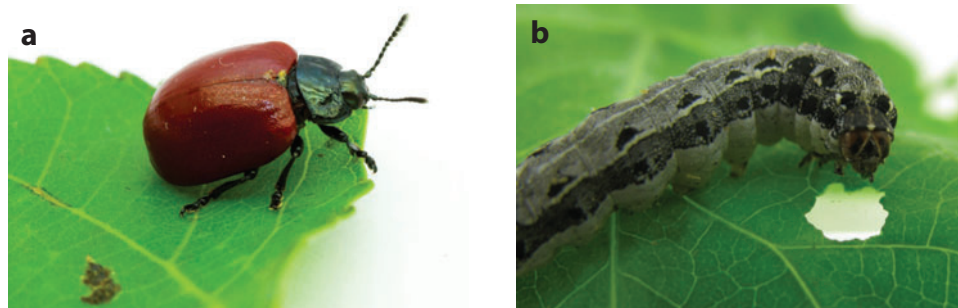


Figure 1

(a) Leaf beetle *Chrysomela populi* feeding on *Populus*. (b) Larva of the generalist lepidopteran herbivore *Spodoptera littoralis* feeding on a lima bean (*Phaseolus lunatus*) leaf.

emission of volatiles has been demonstrated for many different plant species (121), including corn, *Zea mays* (119); cotton, *Gossypium hirsutum* (39, 106); *Lotus japonicus* (94); tobacco, *Nicotiana attenuata* (66); and barrel medic, *Medicago truncatula* (74). Only recently, researchers determined that glandular trichomes can contribute to indirect defense: *Manduca sexta* larvae feeding on *N. attenuata* leaves first take up *O*-acyl sugars present in glandular trichomes. These compounds had been described for *Nicotiana* (6), but the new finding was that, as a consequence of feeding, volatile, branched-chain aliphatic acids released from the *O*-acyl sugars dominate the headspace of the larvae and attract omnivorous ants that attack the herbivore (123).

Many specifications of defenses that are directed against herbivores are present not only aboveground in the green parts of the plants, but also belowground in the rhizosphere (18). This, strikingly, includes VOCs such as (*E*)- β -caryophyllene, which attracts carnivorous nematodes to beetle larvae-infested

maize roots (101), although the distribution of VOCs in soil is strongly limited owing to their adsorption to certain soil particles such as clay. Consequently, VOCs can be considered as infochemicals that mediate various interactions of plants with other species both above- and belowground (18). Finally, providing extrafloral nectar or food bodies is another strategy of indirect defense used by many plants to attract ants, which in turn attack and drive off all other animals from their host plants (54, 70).

In this review, we highlight chemical compound-based principles of plant defenses against herbivores. We discuss their often-disregarded modes of action as well as the arms race between plants and herbivores. Moreover, we consider the impact of additional biotic and abiotic interactions on the plasticity of herbivore-induced chemical defense and use our conclusions to suggest strategies for plant protection.

CHEMICAL DEFENSES: MODES OF ACTION

Because plants can produce a nearly inexhaustible number of metabolites, they possess an enormous reservoir of potentially defensive compounds, many of which have been described in the context of plant interactions with other organisms. These compounds belong to various chemical classes such as isoprene-derived terpenoids including mono-, sesqui-, di-, and triterpenoids as well as steroids;

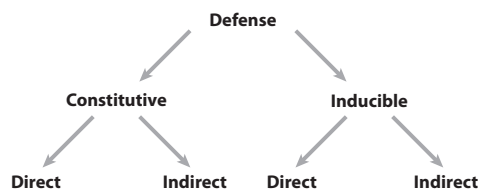


Figure 2

Types of plant defenses.

Extrafloral nectar: nonfloral nectar provided by plants, often involved in the attraction of ants

Table 1 Plants' specialized compounds

Compounds	Example	Typical plant source	Approximate number of compounds known
Terpenoids	(<i>E</i>)- β -Farnesene	Ubiquitous	>30,000
Steroids	Phytoecdyson	Ranunculaceae	~200
Cardenolides	Digoxigenin	Plantaginaceae	~200
Alkaloids	Nicotine	Solanaceae	>12,000
Fatty acid derivatives	(3 <i>Z</i>)-Hexenylacetate	Ubiquitous	n.d.
Glucosinolates	Sinigrin	Capparales	~150
Cyanogenic glucosides	Dhurrin	Rosaceae, Fabaceae	~60
Phenolics	Lignin, tannin	Ubiquitous	>9,000
Polypeptides	Trypsin inhibitor	Ubiquitous	n.d.
Nonprotein amino acids	γ -Aminobutyric acid	Fabaceae	>200
Silica	SiO ₂	Poaceae	1
Latex	Undefined emulsion	Euphorbiaceae	v.c.

Abbreviations: n.d., not determined; v.c., various compositions.

N-containing alkaloids; phenolic compounds including flavonoids; and others (Table 1). These compounds also differ in their structures (Figure 3) (for biosynthetic pathways see related literature), indicating the presence of different target structures. In addition, some compounds occur ubiquitously, whereas others are restricted to certain taxa, for example,

cocaine is specific to the genus *Erythroxylum*, suggesting either broad bioactivity or functions in particular interactions. To minimize the risk of self-intoxication, many defense compounds are usually stored in compartments of limited metabolic activity, such as the vacuole or the apoplast. This is obvious for alkaloids as well as phenolic substances.

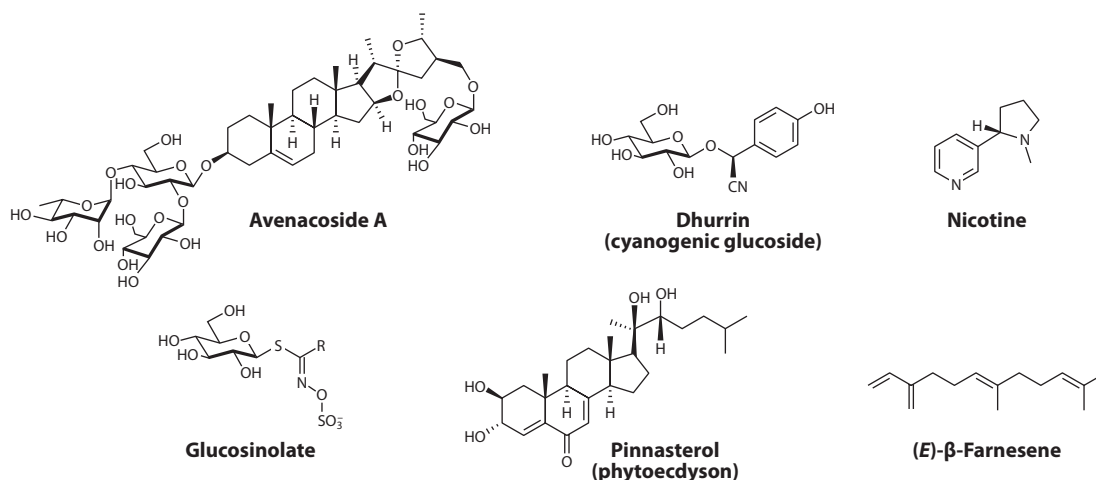


Figure 3

Structures of selected plant defense compounds from various chemical classes: avenacoside A, dhurrin (cyanogenic glucoside), nicotine, glucosinolate, pinnasterol (phytoecdyson), and (*E*)- β -farnesene.

In contrast to the large number of specialized compounds whose involvement in plant defenses against herbivorous insects and other arthropods is known, the exact mode of action on a molecular level as well as the related target structures of these compounds are still unknown. As a result, not all the different compounds or classes of compounds mentioned in the Introduction can be discussed in detail, but some case studies are addressed in the following.

In general, the mode of action frequently includes membrane disruption, inhibition of nutrient and ion transport, inhibition of signal transduction processes, inhibition of metabolism, or the disruption of hormonal control of physiological processes (85, 90, 127). Saponins such as avenacosides (**Figure 3**) have an amphiphilic character and can disrupt cellular membranes (93). Cardenolides (cardiac glycosides) are specific inhibitors of the Na^+/K^+ -ATPase that maintains the electric potential in animal cells, from human to *Drosophila* (81). Cicutoxin, a polyacetylene, prolongs the repolarization phase of neuronal action potentials, very likely by blocking voltage-dependent potassium channels (129). Phytoecdysteroids (**Figure 3**) represent a group of plant compounds that mimic insect hormones, ecdysteroids (including ecdyson), and interfere with the regulation of the periodical molting process (40).

Also discussed are the nonprotein amino acids as defense compounds. In particular, we focus on compounds that show structural similarities with or that are identical to neurotransmitters such as γ -amino butyric acid, GABA, and, thus, can interact with animals' neuroreceptors (60). Interestingly, inorganic compounds can also have a function in defense, e.g., calcium oxalate crystals in *Medicago truncatula* (69) or selenium, as evidenced by the increased protection of hyperaccumulating plants to herbivores (97). Silica, SiO_2 , provides another example of defenses based on inorganic compounds. When included in plant cell walls or when present as silica bodies, it affects food intake by accelerating mandibular wear,

particularly in the case of small insects, and the digestion of plant tissue (29, 104). However, when animals feed mainly on such plants, their teeth wear down more quickly. This is known for some grasses in the African savannas where the incorporation of SiO_2 is inducible under herbivore pressure (83, 84). Generally in terms of direct defenses, most principles of biological activities that make a plant's defense compounds effective against invertebrates are also valid for vertebrate herbivores. Western gray kangaroos avoid feeding on essential oil-containing Myrtaceae (62), formylated phloroglucinol compounds from certain eucalyptus trees act as deterrents to koalas (88), and acacia trees produce higher concentrations of cyanide upon giraffe browsing (136).

Cyanogenic Glycosides

Many constitutively present defensive compounds are noxious or toxic to the plant. Thus, plants must be able to generate and store these substances without poisoning themselves. To achieve this, a commonly used strategy is to store toxins as inactive conjugates, mainly as glycosides (63), and to keep them separate from activating hydrolases. One example is hydrogen cyanide (HCN), which is released from cyanogenic glycosides (**Figure 3**) and present in many (>2,500) plant species (**Table 1**). Cyanogenic glycosides are not toxic and are stored intracellularly in the vacuole, whereas the related glycosidase is present in the cytoplasm. However, upon cell destruction by a feeding herbivore, cleaving off the aglycone moiety is no longer preventable via separation of the enzyme from the substrate. Subsequently, acetone cyanohydrin is released, which can be converted into HCN and acetone either spontaneously or by a hydroxynitrile lyase (122). HCN affects cellular respiration in general by inhibiting the binding of oxygen to the cytochrome-*c*-oxidase within mitochondria; for animals, approximately $100 \mu\text{mol kg}^{-1}$ is a lethal dose (131). However, as is true for various toxins, the dosage is important, and some insect specialists can

Specialized compounds: a diverse group of compounds that are required for neither development nor reproduction but that have a certain ecological function

tolerate greater levels of cyanogenic glycosides (50). Nevertheless, HCN is one of the most effective plant toxins. Thus, it is also necessary that the plant protect itself during biosynthesis of these compounds. This is accomplished by the formation of a multienzyme complex, a metabolon that improves catalytic efficiency by generating cooperating active sites in close proximity and thereby preventing the release of harmful intermediates (132). The newly generated cyanogenic glycosides are very likely directly stored in the vacuole to avoid any contact with the HCN-releasing glycosidases.

Glucosinolates

Besides cyanogenic glycosides, probably the best-known conjugated defense compounds are the glucosinolates (**Figure 3**) (19, 100, 116, 128), present in Brassicaceae, Capparidaceae, and Tropaeolaceae (**Table 1**). Glucosinolates are compartmentalized and thus protected from their hydrolyzing enzyme, a thioglucosidase myrosinase. In contrast to the glucosinolates, which are found distributed in many plant tissues, myrosinase is localized in scattered cells only. Upon tissue damage, both the enzyme and the glucosinolate substrate come into contact: Unstable aglycones are then released, and they spontaneously can rearrange into various active compounds, mainly nitriles and isothiocyanates (19). The latter compounds are toxic to the larvae of the black vine weevil, *Otiorhynchus sulcatus* (20). In a study showing that larvae of *Trichoplusia ni*, a lepidopteran generalist, avoided *Arabidopsis thaliana* ecotypes that produced isothiocyanates upon glucosinolate hydrolysis and, instead, fed on ecotypes that produced nitriles, the biological activity of isothiocyanates was again clearly displayed (71). Interestingly, certain parasitoids use glucosinolates that are released by feeding herbivores to detect their host (59). In such cases, the glucosinolates have a dual function for the infested plant in direct as well as indirect defense.

In addition to their impact on insects, glucosinolates and their hydrolysis products negatively affect a wide range of herbivores

such as mammals, birds, mollusks, and nematodes (116). The broad range of organisms that are affected by isothiocyanates indicates that a general mechanism of toxicity must be responsible. From a chemical standpoint, isothiocyanates are highly reactive compounds: They are electrophilic and react spontaneously with biological nucleophiles such as $-NH_2$, $-SH$, and $-OH$, i.e., the central electrophilic carbon of isothiocyanates ($R-N = C = S$) undergoes rapid addition reactions. Thus, essential compounds in all living cells, mainly proteins but also nucleic acids, may be randomly and uncontrollably covalently modified and, as a result, inactivated (21). Moreover, the tripeptide glutathione (γ -Glu-Cys-Gly) is an abundant physiological thiol that is involved in many redox-regulated cellular processes. An enzymatic reaction mediated by glutathione S-transferases can conjugate isothiocyanates to glutathione, resulting in a thiocarbamate, thereby potentially disturbing the redox homeostasis (21). Whether this holds true and whether a certain preferred target exists remain to be elucidated.

Terpenoids

Terpenoids also contribute to both direct and indirect defenses. They are an extremely diverse group of carbon-based compounds, all of which derived from five-carbon isoprene units and are ubiquitously distributed (**Table 1**). Isoprene may deter herbivorous insects such as *Manduca sexta* (73) but not *Pieris rapae* and *Plutella xylostella* (76). In contrast, isoprene can also affect the attraction of the parasitic wasp *Diadegma semiclausum*, thereby eroding the plant's indirect defense (76). However, the key players in terpenoid volatiles are represented by mono-, sesqui-, and homoterpenoids, which all significantly contribute to any blend of plant-derived volatiles. In terms of indirect defenses, attracting parasitoids or parasites as well as repelling herbivores are very likely mediated by either the recognition of single volatile compounds or of a specific volatile blend by an insect's particular

olfactory system. These interactions between terpenoids and insect sensory receptors have been suggested (49). Using plants that are infested by feeding herbivores to further investigate the volatile compounds by gas chromatography–electroantennogram (GC-EAG) or gas chromatography–single-cell recording (GC-SCR), respectively, investigators may be able to identify signal compounds that are electrophysiologically active and that may subsequently prove to be active in behavioral assays either in predator attraction or, in a more direct way, as repellents of insect pests. For example, monoterpenoids such as linalool and sesquiterpenes such as (*E*)- β -farnesene (**Figure 3**) can be produced by plants and repel herbivores and aphids, respectively (4, 77, 120). By contrast, the C_{16} -homoterpene 4,8,12-trimethyl-1,3(*E*),7(*E*),11-tridecatetraene may be shown to attract predatory mites in behavioral experiments (32). The attraction of the predatory mite *Phytoseiulus persimilis* to (3*S*)-(*E*)-nerolidol has been well demonstrated (85). In general, the exact mechanisms by which terpenoids directly act on insect pests are not known; processes such as the alkylation of nucleophiles, inhibition of ATP-synthase, interference with insects' molting regulation, or the disturbance of the nervous system are very likely (72). As one example for the latter, there exists pharmacological evidence of inhibition of acetylcholine esterase by α -pinene, limonene or eugenol (78).

In addition to their interactions with insects, terpenoids also interfere with other plants. Certain monoterpenes, such as carvacrol and *D*-limonene, serve an allelopathic role by inhibiting respiration, blocking the nitrogen cycle, or inhibiting growth and seed germination of neighboring plants (78). Moreover, plants not only emit volatiles, but also perceive or recognize them in inter- and intraplant communications. Unfortunately, the mode and mechanisms underlying volatile recognition are completely unknown, but receptor-mediated signaling is very likely (85).

All these terpenoid compounds are also main constituents of plant resins, which are present

mainly in conifers, where nonvolatile diterpenoids can also be found (95), thus redounding to the direct defense strategy. In resin, the so-called turpentine fraction of conifer oleoresin includes mono- and sesquiterpenes, which often act as repellents or deterrents (99). In addition, turpentine fraction serves as a solvent to mobilize the diterpenoid resin acids to wounded sites. After volatilizing of the turpentine fraction occurs, the remaining resin acids undergo oxidative polymerization, thereby entrapping and killing invading insects (95).

Particularly for the terpenoids more than for other defensive compounds, the following question remains: Is the insect always the targeted organism? The entire microbial-gut community, which is responsible for food digestion, may be affected by plant-derived defense compounds that are absorbed during the feeding process. Because terpenoids can have antimicrobial activities (8), any negative effect on the composition and function of the bacteria in the gut could lead to drastic consequences for the animal, although the insect was not the original target.

Alkaloids: Nicotine and Others

Alkaloids, in general, are a structurally diverse group of nitrogen-containing basic natural products consisting of more than 20 different classes, e.g., pyrrolidines, tropanes, piperidines, pyridines. Typically, they do not have a primary function in plants, but many are toxic to animals, vertebrates as well as arthropods. Alkaloids act on various metabolic systems in animals; some can affect enzymes and, thus, alter different physiological processes; some intercalate with nucleic acids, thereby inhibiting DNA synthesis and repair; and others have strong effects on the nervous systems. Interestingly, many alkaloids possess multiple functions (125). Typical alkaloids are represented by the tropolone alkaloid colchicine, the purine alkaloid caffeine, the isoquinoline alkaloid sanguinarine, the indolizidine swainsonine, and the pyridine alkaloid nicotine (**Figure 3**). Alkaloid-rich plant families are Solanaceae, Papaveraceae, Apocynaceae, and Ranunculaceae.

Allelopathy: a phenomenon by which a plant produces compounds that affect the growth, survival, and reproduction of other organisms

Colchicine is produced by *Colchicum autumnale*. It inhibits polymerization of microtubules by binding to tubulin, thus inhibiting mitosis, and is toxic (EC_{50}) to *Apis mellifera*, honey bee, at a concentration of 0.03% (w/v) provided with food (35). Sugar-mimicking alkaloids, referred to as imino sugars, represent efficient inhibitors of various glycosidases and sugar-metabolizing enzymes (7, 115). Their toxicity and growth-retardation properties in insects rely on the inhibition of sucrase in the midgut and trehalase in various other tissues, causing the inability to uptake sucrose and utilize trehalose (57). The trihydroxyindolizidine alkaloid, swainsonine, from *Swainsonia canescens* and other legumes is an efficient inhibitor of α -mannosidase (28). Interestingly, in some plant species, swainsonine is synthesized by an endophytic fungus (12). Caffeine is found in various plant species, the most prominent of which is *Coffea arabica*, where it acts as a natural defense compound. Caffeine paralyzes and can be toxic to insects feeding on the plant (EC_{50} : 0.2%; *A. mellifera*) (35). The effect is mainly due to the inhibition of phosphodiesterase activity and to the concomitant increase of the intracellular cyclic AMP level (91). Owing to its interaction with adenosine receptors of the nervous system in vertebrates, caffeine has a stimulating effect,

which may be the reason behind the cultivation of *C. arabica* for thousands of years.

Sanguinarine from *Sanguinaria canadensis* is one example for an alkaloid exhibiting multiple effects: It affects neurotransmission by inhibiting the choline acetyl transferase, DNA synthesis, and also various neuroreceptors (125). Alkaloids can bind to various neuroreceptors and either block or displace the endogenous neurotransmitters, thus acting as agonists or antagonists. Alkaloids often derive from the same biogenic precursor as neurotransmitters and mimic them structurally. One of the best-studied examples is nicotine. As outlined in **Figure 4**, (*S*)-nicotine is assembled in the roots of tobacco plants by the nicotine synthase from the *N*-methyl- Δ^1 -pyrrolinium cation and nicotinic acid (65). The methyl- Δ^1 -pyrrolinium cation itself is derived from putrescine, which also serves as a building block for other tropane alkaloids; it is produced from L-ornithine or L-arginine by specific decarboxylases followed by methylation catalyzed by the putrescine *N*-methyltransferase (PMT) and oxidation to 4-methylbutanal by the diamine oxidase (DAO). 4-Methylbutanal is unstable and cyclizes spontaneously to the 1-methyl- Δ^1 -pyrrolinium cation. The biosynthetic sequence to nicotine is triggered by herbivory,

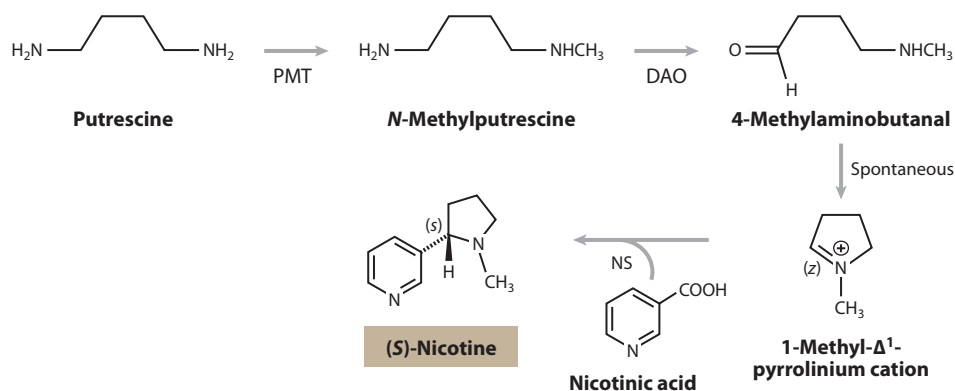


Figure 4

Biosynthesis of nicotine. Nicotine is assembled by condensation of an intermediate in the NAD salvage pathway and the methylpyrrolinium cation derived from ornithine via putrescine. Enzymes involved in nicotine synthesis are indicated: PMT, putrescine *N*-methyl transferase; DAO, diamine oxidase; NS, nicotine synthase.

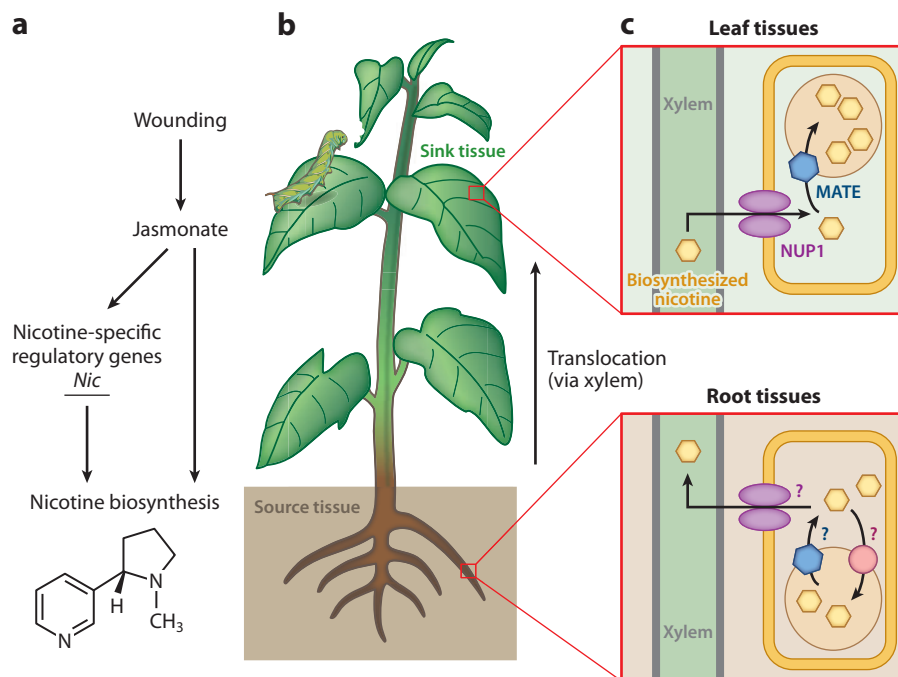


Figure 5

Model for herbivore/wounding-mediated nicotine accumulation in the leaves of *Nicotiana tabacum*. (a) The herbivory-induced phytohormone jasmonate is transported by the phloem to the roots and triggers nicotine biosynthesis along with upregulation of the required transporters. (b) Illustration of the transport routes with transporters in tobacco plants. (c) The biosynthesized nicotine (yellow hexagons) is loaded via a jasmonic acid-induced multidrug and toxic compound extrusion (MATE) transporter into the vacuole. In the leaves a second transporter, nicotine uptake permease (NUP1), translocates the alkaloid from the xylem to the leaf cells. The three different transporters in the roots are still not identified, as indicated by the question marks. Modified after Reference 131.

which results in an enhanced jasmonate level in the wounded leaves (Figure 5). The phytohormone (externally added methyl jasmonate is also active) is transported into the roots and activates the nicotine biosynthesis along with jasmonate-inducible transporters belonging to the tonoplast-localized family of multidrug and toxic compound extrusion (MATE) transporters (89). This type of transporter functions as a proton antiporter and also translocates other alkaloids such as anabasine, hyoscamine, and berberine. The root-produced alkaloid is translocated via the xylem to the aerial parts of the plant. Another transporter, called nicotine uptake permease (NUP1), localized in the plasma membrane allows the alkaloid (and

others) to enter the leaf cells (55). Finally, nicotine is deposited in the vacuoles of the tobacco plant leaves with the help of a MATE transporter, jasmonate-inducible alkaloid transporter 1 (Nt-JAT1) (Figure 5) (89). Other transport proteins still remain to be identified; however, ATP-binding cassette (ABC)-type transporters are promising candidates (131). Nicotine is a long-known defense compound (112) (EC_{50} : 0.2%; *A. mellifera*) (35). Its targets are the nicotinic acetylcholine receptors (nAChRs), the most abundant excitatory postsynaptic receptors in insects (108). In early studies employing electrophysiology and radioligand-binding techniques, researchers identified insect nAChRs as the most likely site

Laticifers: single cells or a group of connected cells containing latex

for nicotine action (48). Resistance to nicotine has been reported in the aphids *Myzus persicae*, *M. nicotianae* (36), and *Aphis craccivora* (37).

Proteinase Inhibitors

Recent studies using microarrays and proteomic approaches revealed that the role of protein-based defense in plants' resistance against herbivores very likely has been underestimated (45, 135). Defense-related proteins such as arginases, ascorbate oxidases, lipoxygenases, polyphenol oxidases, and peroxidases may have antinutritional properties; others such as chitinases, cysteine proteases, lectins, and leucine aminopeptidases may be toxic (135). Many of these proteins are active in the insect gut, given that they can survive the alkaline gut conditions. However, anti-insect activity of toxic plant proteins is easily diminished by proteolyses. Thus, proteolysis-susceptible proteins can be protected by simultaneously providing protease inhibitors (PIs). PIs bind to proteases and inhibit their enzymatic activities. These prevent degradation of the antinutritional or toxic proteins and allow them to exert their function (5). In addition, PIs can affect digestion in the insect gut and, hence, interfere with nutrient utilization. PIs are inducible by insect feeding (51), and their defensive roles against herbivores are well established in many plants (107, 135). For example, herbivore attack on *N. attenuata* rapidly increases the production and accumulation of trypsin PIs; *M. sexta* as well as *Spodoptera exigua* performed better on trypsin PI-deficient plants compared with wild type (113, 134). In tomato (*Lycopersicon esculentum*), PIs were positively tested for their trypsin- and *Helicoverpa armigera* gut proteinase-inhibitory activity in different organs of the plant. Observation in the field also revealed that *H. armigera* larvae infested leaves and fruits but not flowers, a fact that could be correlated with the higher levels of PIs in flower tissues (30). Moreover, serine PIs specifically defend *Solanum nigrum* against generalist herbivores (52). In all examples mentioned serine proteases have been addressed, yet plants contain various types of

PIs affecting serine (trypsin, chymotrypsin), cysteine, metallo, and aspartic-proteases (107).

Latex

Latex is the common name for chemically undefined milky suspensions or emulsions of particles in an aqueous fluid, usually held under pressure in living plant cells referred to as laticifers (2). In 1905, Kniep (68) had already suggested a defensive character of latex from Euphorbiaceae. Latex is present in approximately 10% of all plant species and can contain various specialized metabolites and proteins in concentrations that often are much higher than those in leaves. Such compounds are terpenoids such as rubber (*cis*-1,4-polyisoprene), cardenolides, alkaloids such as morphine in *Papaver* species, various proteins such as digestive cysteine proteases in *Carica papaya* and *Ficus* species, and proteinase inhibitors (2). Many of these compounds provide resistance to herbivores, because they are toxic, antinutritive, or simply sticky. This latter effect is the primary function of rubber, entrapping the insect or miring and gluing its mouthparts (42, 43). Both stickiness and the typically white color of latex are due to the rubber particles being dispersed in the fluid. Upon mechanical wounding of laticifers during feeding, latex immediately leaks from the wound site and may come in contact with the herbivore. Many studies have focused on latex as a trait reducing herbivory or the preference or performance of insect herbivores. For instance, as shown for the milkweed, *Hoodia gordonii*, both larval feeding and adult oviposition by *T. ni* was deterred when latex was added to an artificial diet or painted on the leaves of the host plant (26). However, aside from stickiness, the active compound targeting the herbivore is often unknown because latex is such a rich mixture of many compounds. Notably, herbivores try to avoid contact with latex, and some specialists are able to disarm the latex defense by employing a vein-cutting or -trenching behavior, which severs the laticifers and drains the latex in response

to internal pressure so that insects can start feeding on a distant part of the plant tissue (43).

Most of the available literature shows that a certain insect is affected by constitutive or induced chemical defense of the host plant. Although most studies also demonstrate the important role of the particular defense compounds, the mode of action in these compounds and their target in the enzyme on a molecular level are not known. Thus, the exact modes of action of many of the specialized compounds, which are employed in plant defense, need to be elucidated. This holds true, in particular, for the large classes of terpenoids and alkaloids. In both cases, their activities very likely do not depend on spontaneous and mainly nontargeted reactions with important macromolecules such as proteins. Terpenoids and alkaloids often interact with specific targets, e.g., receptors or certain enzymes, thereby interfering with particular cellular pathways in the insect. It will be interesting to identify such compounds and their corresponding targets to yield a basic structure and further develop highly specific and directed compounds, which could be used for plant protection against insect pests. In addition, it is tempting to speculate that in the reservoir of peptide-based compounds, researchers will find not only PIs, but also inhibitors for other enzymatic activities essential for food digestion, such as have been identified for glycosidases, lipases, or other hydrolytic enzymes.

A certain problem in the identification of active defense compounds may be a result of the uncertainty as to whether a single compound represents the active one or a mix of various compounds, acting on the insect additively as well as synergistically. A better understanding of the underlying mechanisms could open a door into the development of new defense strategies of insects and probably other aggressors.

METABOLIC PLASTICITY AND RESOURCE ALLOCATION

Without doubt, a plant's need to invest in defenses is costly regardless of whether the

defense is constitutive or inducible. The costs are different with respect to the compounds synthesized, e.g., phenolics are suggested to be cheaper than alkaloids because of the additional effort required for inorganic nitrogen to be made bioavailable (11, 25). The defense costs are paid mainly in the form of energy, carbon, and nitrogen. However, their use in defense precludes their availability for growth and reproduction. Calculating such costs is not easy, and several models have been suggested (11). For example, using data from Coley (27) on the neotropical tree *Cecropia palata*, Zangerl & Bazzaz (133) estimated that the allocation of 6% of leaf biomass equivalents to defense caused a 33% reduced growth after 18 months.

As an alternative or additional strategy to the production of defensive compounds, plants can develop a tolerance to herbivory by mobilizing and saving stored energy; an example of this is the allocation of sugars from infested green parts into the nonaffected roots, as has been shown for *Manduca sexta*-infested *Nicotiana attenuata* plants using ^{11}C -labeled photosynthates (109). Thus, at the necessary time, all rescued material can easily be remobilized and used for building new aboveground organs. In this particular case, the delivery of energy and other resources into the roots can also strongly support the generation of nicotine as a defensive compound in *N. attenuata* because its biosynthesis is restricted to the root tissue. Generally, the efficiency and dynamics of all such processes depend on various parameters such as (a) the types of compounds that have been generated and, thus, the availability and interconvertability of the related biosynthetic precursors needed; (b) the pathways involved and their current enzymatic equipment; and (c) the spatial distribution of compounds within plant tissues and organs.

Precise coordination of plant activities and the adaptive modulation of the plant's metabolism can be realized only if plants recognize signals containing information about their direct environment quickly and efficiently, which includes the challenge by herbivores. Upon signal perception, within and between

Polyphenism:

occurrence of various, discrete phenotypes which can emerge from a single genotype resulting from differing environmental conditions

Jasmonic acid-isoleucine conjugate (JA-Ile):

the active form of jasmonates

Jasmonate ZIM-domain protein1 (JAZ1):

repressor of jasmonate-responsive genes

plant cells, this information will then be transduced to a defined sequence of messenger molecules, eventually leading to gene activation and finally the induction of the required plant response (79, 80). Metabolic plasticity of chemical defenses against herbivores can depend, to some extent, on the presence of other environmental cues and, thus, may resemble the phenomenon of polyphenism. A polyphenic trait is a feature for which numerous, discrete phenotypes can emerge on the basis of a single genotype as a result of differing environmental circumstances.

Plant hormones have a key role as mediators in transduction chains and are involved in the regulation of environment-induced plant responses and the expression of the respective metabolic responses, which can show an enormous level of plasticity. However, many developmental processes and adaptive responses are not regulated by one single phytohormone, and induced changes are mediated by sophisticated signaling networks (118, 130). Slight changes in phytohormone concentrations in combination with different tissue sensitivities may cause a range of simultaneous effects because each phytohormone can have several effects. This stresses the importance of phytohormones as regulators connecting environmental signals and plant responses. The metabolic plasticity is realized in the appearance of specified compounds whose synthesis evolved in plants as a result of selection for increased fitness via a better adaptation to the local ecological niche of each species (24). Thousands of terpenoids, alkaloids, and phenylpropanoids have been found in the plant kingdom, but each species is capable of synthesizing only a fraction of this metabolic diversity. Metabolic plasticity reflects the evolutionary plasticity with closely related enzymes from different protein families, differing in their product profiles, localization, or the substrates they use (24).

Abiotic factors can also influence the metabolic phenotype. An obvious example is light, which is sensed by the phytochrome system, i.e., determining the ratio of red to far-red parts of sunlight. Phytochrome, in turn,

controls certain phytohormone levels such as auxins and gibberellins, and it is responsible for the reduction of the plant's sensitivity against the defense-related jasmonates (9). In lima bean, the light environment mediated by the phytochrome system modulates the plant's response to jasmonates as well as JA-Ile (jasmonic acid-isoleucine conjugate) biosynthesis, which controls the subsequent extrafloral-nectar secretion (98). In *Lindera benzoin*, herbivores performed better on sun-exposed leaves than on leaves in the shade owing to the higher activities of defense-related proteins in the latter (87). For *A. thaliana*, jasmonate and phytochrome A (phyA) signaling are integrated via the stability of the jasmonate ZIM-domain protein1 (JAZ1), which is involved in the repression of jasmonate-responsive genes. In this study, phyA mutants showed reduced JA-regulated growth inhibition compared with the wild-type control because the degradation of JAZ1 in response to JA treatment or wounding required phyA, indicating that far-red and defense pathways are integrative (105).

Besides abiotic factors, typical biotic cues can also address the metabolic responses of plants infested by herbivores (22, 114). A study using *Arabidopsis* showed that herbivory could induce resistance against certain pathogens (34). Symbiosis with mycorrhizal fungi also affects secondary metabolism including the defensive traits of host plants. In an investigation of the influence of mycorrhization by *Glomus intraradices* on inducible indirect defenses after *Spodoptera* feeding in *M. truncatula*, researchers measured VOCs emission in mycorrhizal and nonmycorrhizal plants. Although the differences observed in volatile emission are only marginal, classification of a distinguishable volatile pattern was possible (75). In another experiment, a mixture of arbuscular mycorrhizal fungi colonizing *Plantago lanceolata* resulted in suppression of the plant's defense induced by herbivory, at least of the volatile compounds (14, 46). However, deeper insights into multiple interactions are hindered by the fact that all organisms involved have an impact on the outcome. Thus, general effects are

difficult to find. However, as stated by Leitner and colleagues (75), the data available to date indicate that even the slightest variation in one of the partners of any interaction may change the overall consequences.

A similar phenomenon is also known for various developmental stages of some plants. For example, young leaves of lima beans possess a higher capacity in cyanogenesis than do older leaves (10). Some birch species, *Betula platyphylla* and *B. ermanii*, contain higher levels of tannins and phenolic compounds in leaves developing early in the year and are better protected against herbivory because main growth rates occur early in the year (82).

EVOLUTION OF CHEMICAL DEFENSES

Organisms never exist alone: They interact with other organisms existing in their environment such as predators, parasites, hosts, or mutualists. As a consequence, they are exposed to natural-selection pressures borne by other organisms and driving evolution. If the evolution of a particular species results in the evolution of a respective counterpart, and vice versa, they are very likely involved in a coevolution process referred to as an arms race (31, 44, 117). For several million years, plants, insects, and their predators have coevolved on the basis of a chemical arms race that includes the employment of refined chemical defense systems by the antagonists. Although this concept is widely accepted, experimental supporting data are limited. The best-studied example (see below) comprises the “invention” of angular furanocoumarins after the plant’s defense by linear furanocoumarins had been overcome (17).

In plant herbivore interactions, specialized herbivores tend to be less affected by the chemical defenses of the host plant than are generalists (1, 3). This is due to an evolutionary adaptation to certain plant chemicals whereby developing mechanisms detoxify, sequester, excrete, or selectively bind plant defense compounds (23, 58, 61, 92, 111, 124). Such

phenomena are successfully realized in the interactions between the larvae of the lepidopteran specialist insect *Pieris rapae* and plants of the Brassicaceae, which are equipped with the glucosinolate/myrosinase defense. Here, a gut protein can direct the hydrolysis reaction toward nitriles instead of isothiocyanates (126). Another strategy to detoxify glucosinolates has been realized in the diamondback moth, *Plutella xylostella*. In that case, a glucosinolate sulfatase from the insect’s gut generates desulfoglucosinolates, thereby outcompeting the myrosinase. Thus, the insect avoids the production of the toxic isothiocyanates and nitriles (102). Studying *Papilio polyxenes* behavior demonstrates that insects adapted to feeding on toxin-containing host plants through diversification of cytochrome P450 monooxygenases, which are involved in detoxification of furanocoumarins (110). For the Apiaceae, the presence of hydroxycoumarins may be an ancestral trait relative to the more toxic linear furanocoumarins, demonstrating that the more complex angular furanocoumarins are the most derived of all three conditions (15, 16). Indeed, Berenbaum & Zangerl (17) showed that variations in the production of furanocoumarins in the Apiaceae plant, *Pastinaca sativa*, were accompanied by variations in the ability of the herbivorous insect to metabolize these compounds. The high levels of matching between plant and insect phenotypes suggested that the genes conferring an ability to exploit hosts may be tightly linked. An additional study showing a positive evolutionary trend concerning incremental diversity and complexity of chemicals also confirmed the coevolution theory (13). On the basis of the volatile analysis of 70 species in the genus *Bursera*, a net accumulation of new compounds was clearly demonstrated in time during species diversification. In some cases, insects use such primarily chemical defenses as a cue to find their host, or they exploit the plant-derived compounds for their own defense against parasitoids and predators (44, 92). For example, *P. xylostella* females are stimulated by glucosinolates to oviposit their eggs on Brassicaceae (103).

Coevolution: the process of reciprocal adaptation among populations or species

Push-pull strategy: a strategy that employs volatile compounds to repel (push) insect pests from the crop and to attract (pull) them into trap crops

Specialized compound-based defense is often the causative factor in examples where specific plant hosts are fully resistant to an attack by certain insect pests. Furthermore, the ability of the respective herbivore to handle those compounds successfully may generate resistance in insects. However, each of these resistance phenomena will be overcome at the moment when one of the protagonists scales up to the next level in the arms race between plant host and herbivore.

APPLICATIONS AND OUTLOOK

Knowledge regarding the presence, efficiency, and mode of action of specialized compounds effective against herbivores is a prerequisite, if we are to make use of such compounds for human benefit. Here, two different areas are of main interest, agriculture and pharmaceutical. In the latter case, alkaloids, flavonoids (e.g., phytoestrogens), and cardenolides are of strong interest for researchers to identify and develop new drugs to treat various kinds of diseases from cancer and HIV infection to heart disease.

However, it must not be forgotten that many of the defensive compounds such as HCN are present in various crops that can be harmful to livestock farming and humans. In these cases, the generation of plants with lower or no content of such compounds is needed. Their production may be achieved by employing either a classical approach via breeding techniques or modern molecular methods providing genetically modified plants. Alternatively, knowing the compounds that are effective in plant defense against herbivores may help to develop new strategies to protect crop plants from insect pests. Again, breeding or bioengineering can generate plants that produce toxins, repellents, or other protecting compounds, thereby strengthening the crop to withstand successfully herbivore attacks. Metabolic engineering of such compounds can be achieved by modifying existing pathways, for instance, by up- or downregulation of selected biosynthetic steps or by modulating metabolite fluxes by inhibiting all the competing pathways to attain

a desired compound. By overexpressing a linalool/nerolidol synthase (FaNES1) from strawberry (*Fragaria x ananassa*) in *Arabidopsis* chloroplasts, researchers were able to show a successful alteration of volatile-mediated direct plant defense. The aphid *Mysus persicae* was repelled by linalool and its derivatives, which were produced in the transgenic plants using choice experiments (4). The direction of FaNES1 to mitochondria resulted in the synthesis of (3*S*)-(*E*)-nerolidol and its metabolite, the homoterpene 4,8-dimethyl-1,3,7-nonatriene (DMNT), owing to the presence of the sesquiterpene precursor farnesyl diphosphate, which attracted the predatory mite *Phytoseiulus persimilis* (64). Other examples of the successful use of such an approach with overexpressing terpenoids or fatty acid derivatives are described elsewhere (85). Alternatively, regulating the production of defense compounds may also be changed by manipulating phytohormone levels, such as those of jasmonates and salicylic acid, which are key regulators of secondary metabolism.

Another strategy is the employment of plants that emit volatiles that can either attract or repel, thereby impacting insect behavior. In a so-called push-pull strategy, repelling plants must be laid as intercrops to protect (push), whereas attracting plants must be laid around the field (pull). Ideally, in addition to repelling the herbivores, the intercrop attracts and conserves the natural enemies of the herbivorous arthropods, thus assuring a continued suppression of the pest. This strategy is far from being a new development. The Incas in the South American Andes used mashua (*Tropaeolum tuberosum*) plants as intercrop to grow and protect their potato plants. More currently, farmers of small land shares in eastern Africa are using this approach to biological pest control to manage cereal stem borers in millet and maize (67). Thus, crop protection for human benefit does not necessarily mean depending on pesticides, but instead employing and exploiting traditional farming strategies based on plant chemistry, which may also be ecologically justified and sustainable.

SUMMARY POINTS

1. Plants possess constitutive as well as inducible defense mechanisms; both can act directly or indirectly against the aggressor.
2. In infested plants, herbivorous insects induce various defense strategies. Among them, the chemical defense is very powerful owing to the enormous number of different compounds and their high structural diversification, which implicates a very high number of different targets in the herbivores.
3. Chemical defensive compounds such as HCN or nicotine can be toxic to the herbivore, active as repellents, act on various targets to affect the development or (neuro)physiology of the feeding organisms, or inhibit digestion as do proteinase inhibitors. They also can attract an additional trophic level that attacks the herbivores.
4. Plants can efficiently allocate energy and metabolites from existing fixed forms to needed defensive compounds.
5. Owing to additional interactions with the biotic and abiotic environment, a plant's composition of chemical compounds may be modified, indicating a high ability for metabolic plasticity.
6. Herbivorous insects can overcome the negative effects of plant defensive compounds by employing various strategies, such as detoxification, sequestration, or secretion.
7. The arms race between host plants and herbivores is a driving force for coevolution.

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The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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